

f_1 f_2 f_3 f_4 f_{n-1}
 shape size weight color

'I like the movie' -
 (movie is not good)
 (wasted my time)
 'wasted'

f_1 f_2 f_3 f_4
 positive: like liked good
 negative: not wasted
 neutral: not

① hand drawn
 ② on paper
 ③ from screen

	like	liked	good	not	wasted
movie is not good	1	0	0	0	0
	0	0	1	1	0

'Paraphrase'
 ['_____', '_____', '_____']

① sentence tokenizer, ['the', 'was']

- ✓ ② word tokenizer
- ✓ ③ remove stopwords
- ✓ ④ remove punctuation (extra)
- ⑤ Stemming
- ✓ ⑥ lemmatize
- ⑦ embeddings - numerical feature

① —
 ② —
 ③ —

Corpus = [like movie good wasted time liked]
Count vector

Corpus = [like movie good wasted time like]

'i like the movie'		1	1	0	0	0	0
movie is not <u>good</u>		0	1	1	0	0	0
wasted my time		0	0	0	1	1	0
like them		0	1	0	0	0	1



- ① word (tree)
 - ② stop word
 - ③ feature
 - ④ lemma
 - ⑤ Count Vector
- x = y =
- ⑥ apply

Corpus = []

docs = ['the boy and the dog' → d₁
 'the cat and the dog' → d₂]

minimise vocabulary = ['and', 'boy', 'cat', 'dog', 'the']

universe vocabulary = ['and', 'boy', 'cat', 'dog', 'the']

total terms = 5 (N=5)

<u>term</u>	count	<u>TF calc</u>	<u>TFIDF</u> TF value
		1/5	0.2
and	1		
boy	1	1/5	0.2
cat	0	0/5	0
dog	1	1/5	0.2
the	2	2/5	0.4

the cat and the dog

dr	term	TF	value
		1/5	0.2
	and		0
	boy	0/5	
		1/5	0.2
	cat	1	0.2
	dog	1/5	0.2
	the	2/5	0.4

$$IDF(t) = \log\left(\frac{N_{docs}}{N_{docs}(t)}\right)$$

Ndocs = 2

the boy and the

	<u>docs calc</u>	IDF	the cat and the
and	<u>2</u>	$\log(2/2)$	0
<u>boy</u>	1	$\log(2/1)$	0.69
cat	1	$\log(2/1)$	0.69

dog	2	ws (42)	0
the	2	ws (42)	0

TF-IDF

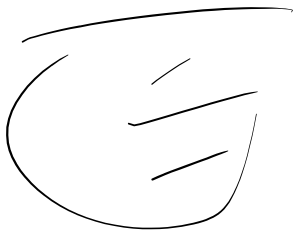
doc	TF	IDF
0.2	0	
0.2	0.69	
0	0.69	
0.2	0	
0.4	0	

TF-IDF

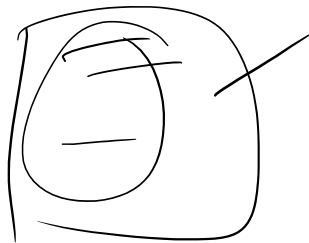
0
0.138
0
0
0

doc [0 0.138 0 0 0]

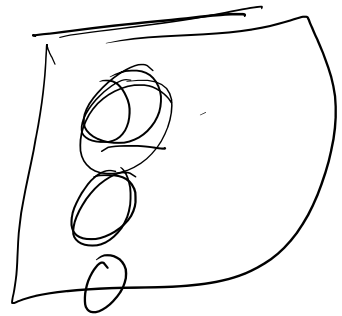
Web scraping



may



2



<table>		
th		
<tr>	name	ale
<tr>	id	2

<table> (all 2 <table> ... name / rth <th> ale </th> </tr>)

